

Sangana Venkata Sai Kiran

Hyderabad, India | venkatasaikiran71@gmail.com | +91-75693 32604 | [LinkedIn](#)

PROFESSIONAL SUMMARY

Java Full Stack Developer with **3.5+ years** of experience designing and delivering scalable enterprise applications and event-driven microservices for **American Airlines**. Expertise in building distributed systems with **Spring Boot, Apache Kafka, and Microservices Architecture**, processing **100K+ events daily** with sub-second latency. Skilled in modernizing legacy systems on **Azure Kubernetes Service (AKS)** and crafting responsive UIs with **Angular and TypeScript**. Proven ability to reduce latency by 60%, achieve 99.9% reliability, and deliver business-critical features in agile environments.

TECHNICAL SKILLS

Languages: Java 8/11/17, TypeScript, JavaScript, SQL

Backend: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, RESTful APIs, Microservices Architecture

Frontend: Angular 12+, RxJS, HTML5, CSS3, Bootstrap, Reactive Forms

Event-Driven & Messaging: Apache Kafka, Spring Kafka, Event-Driven Architecture, Pub/Sub Patterns

Databases: MySQL, SQL Server, JPA/Hibernate

Cloud & DevOps: Azure, Azure Kubernetes Service (AKS), Docker, Kubernetes, GitHub Actions, Rancher

Security: OAuth2, JWT, Spring Security, Azure AD (AAD), Apigee API Gateway, HashiCorp Vault

Monitoring & Tools: Dynatrace, Mezmo, SonarQube, IntelliJ IDEA, Postman, DBeaver, Git, Azure DevOps

Methodologies: Agile/Scrum, SDLC, TDD, Code Reviews, CI/CD

PROFESSIONAL EXPERIENCE

Cognizant Technology Solutions

Apr 2022 – Present

Client: American Airlines (Dallas, TX)

Hyderabad, India

Associate – Backend Engineer (Eventing Platform)

Jan 2025 – Present

- Engineered a real-time **event-driven framework** processing **100K+ flight disruption events daily** with sub-second latency to trigger crew notifications, enabling faster operational recovery during irregular operations.
- Designed and implemented **Kafka-based streaming pipelines** using **Spring Kafka**, reducing notification latency by **60%** compared to legacy batch systems and improving on-time crew dispatch.
- Built fault-tolerant Kafka consumers with **manual acknowledgment, consumer groups, and partition-aware processing** to ensure message ordering and at-least-once delivery semantics.
- Implemented **Dead Letter Topic (DLT)** and retry mechanisms with exponential backoff to ensure **99.9% data reliability** and zero message loss for business-critical events.
- Developed idempotent message handlers using tracking IDs to prevent duplicate event processing across distributed consumers.
- Secured event streams and REST APIs using **Spring Security, OAuth2, JWT, and Azure AD (AAD)** for authentication and role-based authorization.
- Integrated **Mezmo** for centralized logging and **Dynatrace** for APM, reducing issue resolution time by **40%** and improving observability across distributed services.
- Collaborated with cross-functional teams in agile sprints; participated in code reviews, technical design discussions, and on-call rotations.

Programmer Analyst – Full Stack Developer (Communication Systems)

Jan 2024 – Dec 2024

- Modernized legacy communication modules into a **Microservices Architecture** hosted on **Azure Kubernetes Service (AKS)**, managed via Rancher, supporting thousands of pilots and crew across operational hubs.
- Developed responsive **Angular dashboards** for the Admin Portal with reactive forms and RxJS, enabling real-time broadcast notifications and emergency alerts to pilots.
- Designed and implemented **Spring Boot REST APIs** with proper validation, exception handling via **@ControllerAdvice**, and **@Transactional** boundaries for data consistency.
- Secured backend APIs using **Apigee API Gateway** with **OAuth2, JWT tokens, and rate limiting**, mitigating abuse and ensuring compliance.
- Managed application secrets (database credentials, API keys, certificates) securely using **HashiCorp Vault**, eliminating hardcoded credentials.
- Optimized **SQL queries** and database interactions through indexing, JPA fetch strategies, and N+1 query elimination, reducing page load times by **30%** during peak operational windows.
- Containerized services with **Docker** and deployed via **Kubernetes (AKS)** with health checks, rolling deployments, and horizontal pod autoscaling.

Programmer Analyst – Full Stack Developer (Pilot Trip Trade System)

Oct 2022 – Dec 2023

- Delivered core features for the **Pilot Trip Trade System** enabling pilots to swap, drop, and pickup flight sequences, processing **thousands of trade requests** per month.
- Built intuitive **Angular UIs** with reactive forms, RxJS, and component-based architecture for real-time roster updates and trade workflow management.

- Developed RESTful microservices with **Spring Boot** for trade matching, eligibility validation, and audit logging.
- Implemented **role-based access control (RBAC)** and JWT authentication for secure pilot operations.
- Wrote comprehensive unit tests using **JUnit and Mockito**, achieving **80%+ code coverage**.

Full Stack Java Intern

Apr 2022 – Sep 2022

- Built an **Audit Management System** prototype using Java, Spring Boot, and Angular for tracking compliance workflows.
- Gained hands-on experience in **REST API development**, JWT-based authentication, and database design.

KEY ACHIEVEMENTS

- Recipient of **Cognizant Cheers Bronze Award** for outstanding contributions and project deliverables (2024).
- Reduced event notification latency by **60%** through architectural redesign from batch to streaming.
- Achieved **99.9% reliability** for distributed event processing through DLT, retry, and idempotency patterns.
- Mentored junior developers and interns on Spring Boot best practices and code review standards.

EDUCATION

Bachelor of Technology, Computer Science & Engineering

2018 – 2022

Sreenidhi Institute of Science and Technology (SNIST), Hyderabad

CGPA: 7.56/10

CERTIFICATIONS

- **Programming in Java** – NPTEL
- **UI Technologies** – Durgasoft Software Solutions
- Active participant in **Smart India Hackathon** and Competitive Programming (CodeChef, HackerRank)